```
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% 2ECE-A
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```

PROBLEM A: TRANSPOSE OPERATIONS

```
R = [ 5, -2,  0,  7;
    1,  3, -4,  2;
    6,  0,  8, -1 ];


Z = [  2 + 3i, -1 + 4i, 5 - 2i;
    -3 + 1i,       6, 2 + 5i ];
```

Real Matrix displays

```
disp('Real Matrix R');
disp(transpose(R));
disp(R');
disp(R.');


% Complex Matrix displays
disp('Complex Matrix Z');
disp(transpose(Z));
disp(Z');
disp(Z.');
```

a. Which operations give identical results for the real matrix R?

```
% All three operations give identical results
```

b. Which two operations give identical results for the complex matrix Z?

```
% The transpose(Z) function and the dot ' operator Z'
```

c. What additional operation is performed by the apostrophe operator ' when the matrix contains complex entries?

```
% the apostrophe conjugates the complex numbers
```

PROBLEM B: TWO METHODS FOR A LINEAR SYSTEM

```
A = [  2, -1,  3,  2;
```

```
    1,  4, -2,  1;
    3,  2,  1, -1;
    -2,  3,  4,  2 ];


b = [ 16;
    -7;
    6;
    7 ];
```

Inverse Method

```
x_inv = inv(A) * b;
```

Cramer's Rule

```
detA = det(A);
n = length(b);
x_cramer = zeros(n, 1);


for i = 1:n
    Ai = A;
    Ai(:, i) = b;
    x_cramer(i) = det(Ai) / detA;
end
```

Displays and checks

```
disp('Inverse Method Solution (x_inv)');
disp(x_inv);

disp('Cramer''s Rule Solution (x_cramer)');
disp(x_cramer);

disp('det(A)');
disp(detA);

disp('Norm Checks');
disp(norm(x_inv - x_cramer));
disp(norm(A * x_inv - b));
```